

Chapter 4: Network Layer - Data Plane: Easy Study Notes for Exam

This chapter delves into the "data plane" of the network layer, focusing on the practical movement of data packets.

1. Goals of the Network Layer

The network layer's primary role is to move data segments from a sending device to a receiving device across a network.

- **Network Layer Service Models:** Defines the promises the network makes about delivering datagrams.
- **Forwarding vs. Routing:** Differentiating between local packet movement and global path determination.
- **How a Router Works:** Understanding the internal components and processes.
- **Addressing:** How devices are uniquely identified (IP addresses).
- **Generalized Forwarding:** More flexible packet movement methods.
- **Internet Architecture:** The structural overview of the Internet.
- **IP Protocol:** The core protocol of the Internet.
- **NAT, Middleboxes:** Special network functions.

2. Network Layer Functions

The network layer is like the postal service of the internet. Its main job is to take data from one computer and deliver it to another computer anywhere in the world.

The network layer is akin to a postal service, responsible for transporting data "packages" (datagrams).

- **Sender:** Takes data segments, encapsulates them into "datagrams" (network layer packets), and passes them to the link layer.
- **Receiver:** Receives datagrams and extracts the segments, delivering them to the transport layer.
- **Routers:** Key devices that inspect IP datagrams and move them from an input port to an output port, guiding them closer to their destination.

Two Key Network Layer Functions:

- **Forwarding (Data Plane):** This is the **local action** of moving a packet from a router's input link to the correct output link. It's like navigating a single intersection. This is a **per-router function**.
- **Routing (Control Plane):** This is the **network-wide process** of determining the entire path (route) a packet will take from the source to the destination. It's like planning your whole trip from start to finish.

Control Plane Approaches:

- **Traditional Routing Algorithms:** Each router runs these algorithms and communicates with other routers to collectively determine the best paths.
- **Software-Defined Networking (SDN):** A central "remote controller" calculates and distributes forwarding rules (tables) to all routers.

3. Network Layer Service Model

This describes the guarantees the network provides for datagram delivery.

Internet "Best Effort" Service Model:

The Internet offers a "best effort" service, meaning it tries its best but makes **no guarantees** on:

- **Successful delivery:** Datagrams may not always reach their destination.
- **Timing or order:** Datagrams might arrive late or out of order.
- **Bandwidth:** There's no guaranteed data transfer speed.

Why "Best Effort" Works Well:

- **Simplicity:** Its straightforward design facilitated rapid Internet growth.
- **Sufficient Bandwidth:** Often, enough bandwidth is available for real-time applications to function adequately.
- **Distributed Services:** CDNs and data centers deliver content from multiple locations, improving performance.
- **Congestion Control:** Mechanisms like TCP's congestion control manage network traffic effectively.

4. What's Inside a Router

A router has a high-level architecture with two main parts:

- **Forwarding Data Plane (Hardware):** Handles fast packet movement in nanosecond timeframes. It includes:
 - **Input Ports:** Where packets arrive.
 - **Switching Fabric:** Connects input ports to output ports.
 - **Output Ports:** Where packets leave.
- **Routing and Management Control Plane (Software):** Handles routing decisions and network management in millisecond timeframes. It includes:
 - **Routing Processor:** Runs the routing algorithms.

Input Port Functions:

- **Line Termination:** Physical layer reception of bits.
- **Link Layer Protocol:** Handles link-level communication (e.g., Ethernet).

- **Lookup, Forwarding, Queueing:**
 - **Decentralized Switching:** Input ports use a "forwarding table" to decide the output port. This is "match plus action."
 - **Line Speed:** Input port processing aims to keep pace with incoming data.
 - **Input Port Queueing:** Packets wait in queues if the switch fabric is slower than the arrival rate.

Destination-Based Forwarding & Longest Prefix Matching:

- **Destination-based forwarding:** Traditional forwarding relies solely on the destination IP address.
- **Longest Prefix Matching:** When a router searches for a forwarding entry, it selects the entry with the longest matching prefix of the destination address. This is often done using **Ternary Content Addressable Memories (TCAMs)** for high-speed lookups.

Switching Fabrics:

The switching fabric moves packets from an input port to an output port within the router.

- **Switching Rate:** Ideally, for N input ports, the switching rate should be N times the line rate.
- **Types of Switching Fabrics:**
 - **Switching via Memory:** Older, slower method where packets were copied to system memory, limited by memory bandwidth.
 - **Switching via a Bus:** Packets move over a shared bus, limited by bus bandwidth.
 - **Switching via Interconnection Network:** Uses crossbar switches or multi-stage networks for parallelism and higher speeds (e.g., modern Cisco CRS routers can achieve 100s of Tbps).

Router Queueing and Buffer Management:

- **Input Port Queueing:** Occurs if the switch fabric is slower than the combined input port speed, leading to delays and packet loss.
 - **Head-of-the-Line (HOL) Blocking:** A packet at the front of the queue can block other packets behind it, even if those packets could go to a free output port.
- **Output Port Queueing:** Occurs if packets arrive from the switch fabric faster than the output link can transmit them, also leading to delays and potential packet loss.
 - **Buffer Management:**
 - **Drop Policy:** Decides which packets to drop when buffers are full (e.g., tail drop, priority dropping).
 - **Marking:** Packets can be marked to signal congestion (e.g., ECN, RED).
- **How much buffering?** A common rule of thumb is to have buffering equal to the "typical" Round Trip Time (RTT) multiplied by the link capacity. Too much buffering can increase delays.

Packet Scheduling:

This determines which packet to send next from the queue on an output link.

- **First-Come, First-Served (FCFS) / FIFO:** Packets are transmitted in arrival order.
- **Priority Scheduling:** Higher-priority packets are sent first.
- **Round Robin (RR) Scheduling:** Cycles through different class queues, sending one packet from each in turn.
- **Weighted Fair Queuing (WFQ):** An advanced Round Robin variant, providing weighted service to traffic classes for minimum bandwidth guarantees.

5. IP: The Internet Protocol

The IP protocol is central to the network layer in the Internet.

- **IP Datagram Format:** An IP datagram (packet) has several fields:
 - **Version:** IP protocol version (e.g., IPv4, IPv6).
 - **Header Length:** Length of the IP header in bytes.
 - **Type of Service (Diffserv, ECN):** For quality of service.
 - **Total Datagram Length:** Total length of the datagram in bytes.
 - **Identifier, Flags, Fragment Offset:** For fragmentation and reassembly.
 - **Time to Live (TTL):** Maximum hops a datagram can take (decremented at each router).
 - **Upper Layer Protocol:** Indicates the protocol in the payload (e.g., TCP or UDP).
 - **Header Checksum:** For error checking of the header.
 - **Source IP Address:** Sender's IP address.
 - **Destination IP Address:** Receiver's IP address.
 - **Options:** Optional fields (e.g., timestamp, record route).
 - **Payload Data:** The actual data (e.g., a TCP or UDP segment).

Important Numbers to Remember:

- Minimum packet header: 20 bytes
- Maximum packet size: 64,000 bytes
- Typical packet size: 1,500 bytes

6. IP Addressing

- **IP Address:** A 32-bit identifier assigned to each host or router interface.
- **Interface:** A connection between a host/router and a physical link. Routers have multiple interfaces, hosts typically one or two.

- **Dotted-Decimal Notation:** IP addresses are commonly written as four numbers separated by dots (e.g., 223.1.1.1).

Subnets:

- **Definition:** A group of device interfaces that can physically reach each other without passing through a router.
- **IP Address Structure:** IP addresses are divided into two parts:
 - **Subnet Part:** High-order bits common to all devices in the same subnet.
 - **Host Part:** Remaining low-order bits identifying a specific device within the subnet.
- **Subnet Mask:** Indicated by /x (e.g., /24), where x is the number of bits in the subnet part.
- **CIDR (Classless Inter-Domain Routing):** Allows the subnet portion to be of arbitrary length, providing flexible address allocation (e.g., 200.23.16.0/23).

How a Host Gets an IP Address:

- **Hard-coded:** Manually set by a system administrator.
- **DHCP (Dynamic Host Configuration Protocol):** Most common method. A host automatically obtains an IP address from a network server when joining a network.

7. DHCP (Dynamic Host Configuration Protocol)

Goal: A host automatically obtains an IP address from a network server.

- **Address Reuse:** Allows addresses to be reused when a device disconnects.
- **Mobile Users:** Supports users who frequently join and leave networks.

DHCP Process (Discover, Offer, Request, ACK):

1. **DHCP Discover (Optional):** The new host broadcasts a "DHCP discover" message to find a DHCP server.
2. **DHCP Offer (Optional):** A DHCP server responds with a "DHCP offer" message, proposing an IP address.
3. **DHCP Request:** The host sends a "DHCP request" message, asking to use the offered IP address.
4. **DHCP ACK:** The DHCP server sends a "DHCP ACK" (acknowledgement) message, confirming the IP address assignment and its lease time (how long the address is valid).

More than just IP Addresses:

DHCP can also provide other critical information to a client, such as:

- **Address of the first-hop router:** The gateway to the Internet.
- **Name and IP address of the DNS server:** For resolving domain names to IP addresses.

- **Network Mask:** To determine the network and host parts of the address.

This process often involves messages encapsulated in UDP, then IP, then Ethernet, and broadcast on the local network.

8. Internet Fairness (Net Neutrality)

From the US Government, net neutrality broadly implies three simple rules:

1. **No Blocking:** Internet service providers (ISPs) cannot block lawful content, applications, services, or non-harmful devices.
2. **No Throttling:** ISPs cannot impair or degrade lawful Internet traffic on the basis of content, applications, services, or non-harmful devices.
3. **No Paid Prioritization:** ISPs cannot favor some lawful Internet traffic over other lawful traffic in exchange for consideration (i.e., money or other benefits).

Easy Memory Tricks for Exam

- **Forwarding** = Single hop (like one traffic light)
- **Routing** = Complete journey (like GPS planning whole trip)
- **Internet** = "Best effort" (no promises, but cheap and works)
- **Switching Types:**
 - **Memory** = Slow (one person doing everything)
 - **Bus** = Medium (shared highway gets congested)
 - **Network** = Fast (many parallel highways)
- **Scheduling Types:**
 - **FIFO** = Fair but simple
 - **Priority** = VIP gets to cut in line
 - **Round Robin** = Everyone takes turns
 - **WFQ** = Turns based on how much you paid
- **Important Numbers:**
 - IP header: 20 bytes
 - TCP header: 20 bytes
 - DHCP client port: 68
 - DHCP server port: 67
 - Typical lease: 3600 seconds

What to Focus on for Exam

1. Explain the difference between forwarding and routing.
2. Know what "best effort" means and why the Internet uses this model.
3. Understand longest prefix matching and how routers use it.
4. Be familiar with the four scheduling methods (FCFS, Priority, RR, WFQ).
5. Memorize the key fields of the IP packet format.
6. Understand subnets and CIDR notation.
7. Know the 4-step DHCP process.
8. Understand router queuing issues (input/output queuing, HOL blocking) and buffer management.
9. Be aware of the basic principles of net neutrality.

Quick Review Questions to Test Yourself

1. What's the difference between forwarding and routing?
2. Why does the Internet use "best effort" service?
3. What happens when a router's switching fabric is too slow?
4. How does longest prefix matching work?
5. What are the four DHCP messages?
6. What does /24 mean in an IP address like 192.168.1.0/24?
7. What are the three net neutrality rules?
8. Why do we need queuing in routers?
9. What's the difference between input and output queuing?
10. How big is an IP packet header?

Buffer Management

- Buffer management involves deciding which packet to drop when buffers are full.
- Common drop policies include tail drop (dropping the arriving packet) and priority-based dropping/removal.
- Marking packets can signal congestion using techniques like ECN and RED.

Packet Scheduling

- Packet scheduling determines the order in which packets are sent on a link.
- Common scheduling algorithms include first come, first served (FCFS), priority-based scheduling, round robin, and weighted fair queueing.
- FCFS, also known as FIFO, transmits packets in the order they arrive at the output port.

Scheduling Policies: Priority

- Priority scheduling classifies arriving traffic and queues it by class, using header fields for classification.
- Packets are sent from the highest priority queue that has buffered packets, with FCFS within each priority class.

Scheduling Policies: Round Robin

- Round Robin (RR) scheduling classifies arriving traffic and queues it by class.
- The server cyclically scans class queues, sending one complete packet from each class (if available) in turn.

Scheduling Policies: Weighted Fair Queueing

- Weighted Fair Queueing (WFQ) is a generalized form of Round Robin scheduling.
- It provides a minimum bandwidth guarantee per-traffic-class, with each class having a weight that determines its share of service in each cycle.

Sidebar: Network Neutrality

- Network neutrality involves how an ISP should share/allocate its resources, including packet scheduling and buffer management.
- It encompasses social and economic principles such as protecting free speech and encouraging innovation.
- Different countries have different approaches to network neutrality, which are enforced through legal rules and policies.

Sidebar: Network Neutrality

- The 2015 US FCC Order on Protecting and Promoting an Open Internet includes three key rules.
- No blocking of lawful content, applications, services, or non-harmful devices.
- No throttling or impairing/degrading lawful Internet traffic based on content, application, or service.
- No paid prioritization.

ISP: Telecommunications or Information Service?

- The US Telecommunication Act of 1934 and 1996 defines Title II as imposing "common carrier duties" and Title I applying to information services.
- Title II includes reasonable rates and non-discrimination, requiring regulation, while Title I has no common carrier duties but grants FCC authority.
- Whether an ISP is a telecommunications service or an information service provider is critical from a regulatory standpoint.

Network Layer: "Data Plane" Roadmap

- The network layer involves the data plane and control plane.
- Key components include routers with input ports, switching mechanisms, and output ports, along with buffer management and scheduling.
- Topics covered also include IP, generalized forwarding, SDN, and middleboxes.

Network Layer: Internet

- Host and router network layer functions include the IP protocol for datagram formatting, addressing, and packet handling.
- The ICMP protocol is used for error reporting and router signaling.
- Path-selection algorithms are implemented in routing protocols and SDN controllers.

IP Datagram Format

- The IP datagram format includes fields for version, header length, type of service, total length, identification, flags, fragment offset, TTL, upper layer protocol, header checksum, source IP address, destination IP address, and options.
- Key fields include TTL, which is decremented at each router, and source/destination IP addresses.
- The header checksum ensures integrity.

IP Addressing: Introduction

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IP Addressing: Introduction

- Interfaces are connected via wired Ethernet interfaces connected by Ethernet switches, or wireless WiFi interfaces connected by a WiFi base station.
- The document does not delve into how one interface connects to another without an intervening router at this point.

Subnets

- Subnets are device interfaces that can physically reach each other without passing through an intervening router.
- IP addresses have structure, with a **subnet part (common high order bits)** and a host part (remaining low order bits).

Subnets

- To define subnets, detach each interface from its host or router, creating isolated networks.
- Each isolated network is called a subnet.
- **A subnet mask (e.g., /24) indicates the number of high-order bits that represent the subnet part of the IP address.**

Subnets

- The document provides a network diagram and asks to identify the subnets.
- It also asks to identify the /24 subnet addresses.

IP Addressing: CIDR

- CIDR (Classless InterDomain Routing) allows the subnet portion of an address to be of arbitrary length.
- The address format is a.b.c.d/x, where x is the number of bits in the subnet portion.

IP Addresses: How to Get One?

- There are two questions: how a host gets an IP address within its network (host part) and how a network gets an IP address for itself (network part).

- Hosts can get IP addresses either hard-coded by a sysadmin or dynamically via DHCP.

DHCP: Dynamic Host Configuration Protocol

- DHCP allows a host to dynamically obtain an IP address from a network server when it joins the network and renew its lease on the address.
- DHCP supports address reuse and mobile users.
- The DHCP process involves a discover message, an offer message, a request message, and an ACK message.

DHCP Client-Server Scenario

- A DHCP server is often co-located in a router, serving all subnets to which the router is attached.
- An arriving DHCP client needs an address in this network.

DHCP Client-Server Scenario

- The DHCP process starts with the client broadcasting a DHCP discover message to find a DHCP server.
- The server responds with a DHCP offer message, proposing an IP address.
- The client requests the offered IP address with a DHCP request, and the server confirms with a DHCP ACK message.

DHCP: More Than IP Addresses

- DHCP can provide more than just an IP address, including the address of the first-hop router, the name and IP address of a DNS server, and the network mask.

DHCP: Example

- A connecting laptop uses DHCP to get its IP address, the address of its first-hop router, and the address of a DNS server.
- The DHCP REQUEST message is encapsulated in UDP, IP, and Ethernet frames, and broadcast on the LAN.

DHCP: Example

- The DHCP server formulates a DHCP ACK containing the client's IP address, first-hop router IP address, and DNS server information.
- The encapsulated DHCP server reply is forwarded to the client, which demuxes the message up to the DHCP layer.
- The client now knows its IP address, DNS server information, and first-hop router IP address.

Switching via a Bus

- Datagrams are transferred from the input port memory to the output port memory via a shared bus.
- Bus contention limits the switching speed due to bandwidth constraints.

Switching via Interconnection Network

- Interconnection networks like crossbars and Clos networks are used to connect processors in multiprocessor systems.
- Multistage switches are built from multiple stages of smaller switches, exploiting parallelism.
- Datagrams can be fragmented into fixed-length cells, switched through the fabric, and reassembled at the exit.

Switching via Interconnection Network

- Scaling is achieved by using multiple switching planes in parallel to increase speed and capacity.
- The Cisco CRS router uses multiple switching planes with a 3-stage interconnection network to achieve high switching capacity.

Input Port Queuing

- If the switch fabric is slower than the combined input ports, queueing occurs at the input queues, causing delay and loss.
- Head-of-the-Line (HOL) blocking occurs when a queued datagram at the front of the queue prevents others from moving forward.

Output Port Queuing

- Buffering is needed when datagrams arrive from the fabric faster than the link transmission rate.
- The drop policy determines which datagrams to drop if there are no free buffers.
- Scheduling discipline chooses among queued datagrams for transmission, affecting performance and network neutrality.

Output Port Queuing

- Buffering occurs when the arrival rate via the switch exceeds the output line speed.
- Queueing and loss can occur due to output port buffer overflow.

How Much Buffering?

- A rule of thumb suggests average buffering equal to the typical RTT times the link capacity.

- However, too much buffering can increase delays, especially in home routers, leading to poor performance for realtime apps and sluggish TCP response.
- A more recent recommendation is that with N flows, buffering should equal $RTT * C / N$.